

9. Property Data Analysis Using Pandas

```
import pandas as pd
property_data = pd.DataFrame({
    "Property_ID": [101, 102, 103, 104, 105],
    "Location": ["Hyderabad", "Chennai", "Hyderabad", "Bangalore", "Chennai"],
    "Bedrooms": [3, 5, 4, 6, 2],
    "Area_sqft": [1500, 2500, 1800, 3000, 1200],
    "Listing_Price": [5000000, 8500000, 6200000, 12000000, 4200000]
})
print("Property Data")
print(property_data)
avg_price = property_data.groupby("Location")["Listing_Price"].mean()
print("\nAverage Listing Price by Location")
print(avg_price)
count_bedrooms = property_data[property_data["Bedrooms"] > 4].shape[0]
print("\nNumber of Properties with More Than 4 Bedrooms:", count_bedrooms)
largest_area = property_data.loc[property_data["Area_sqft"].idxmax()]
print("\nProperty with Largest Area")
print(largest_area)
```

```
Property Data
  Property_ID  Location  Bedrooms  Area_sqft  Listing_Price
0         101  Hyderabad         3        1500         5000000
1         102   Chennai         5        2500         8500000
2         103  Hyderabad         4        1800         6200000
3         104  Bangalore         6        3000        12000000
4         105   Chennai         2        1200         4200000
```

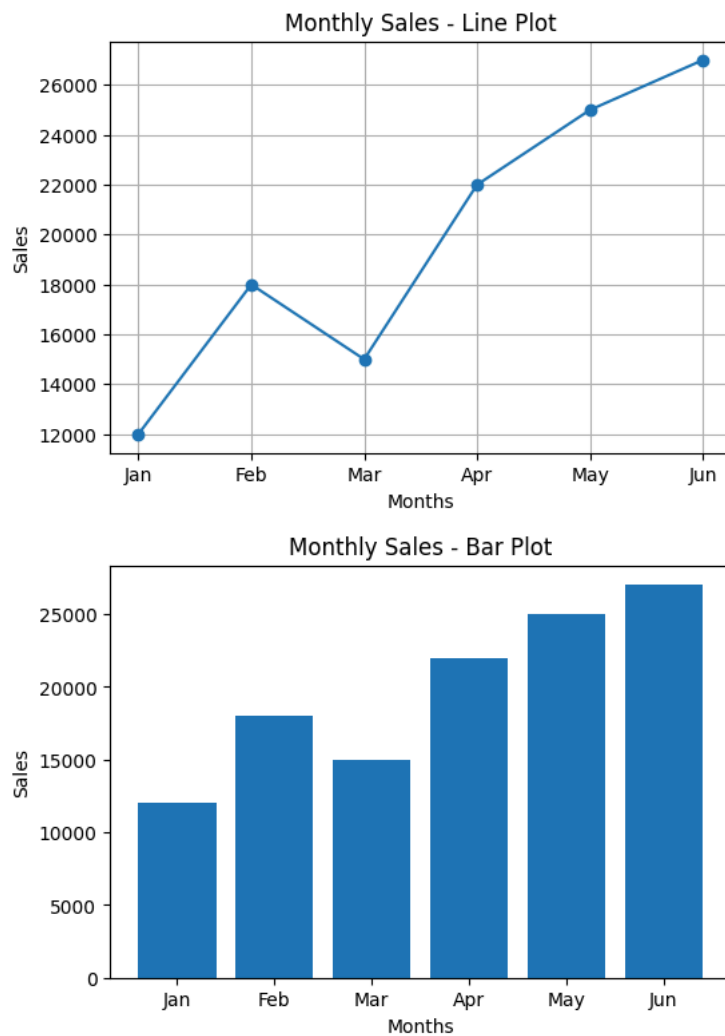
```
Average Listing Price by Location
Location
Bangalore    12000000.0
Chennai       6350000.0
Hyderabad     5600000.0
Name: Listing_Price, dtype: float64
```

```
Number of Properties with More Than 4 Bedrooms: 2
```

```
Property with Largest Area
Property_ID      104
Location         Bangalore
Bedrooms          6
Area_sqft         3000
Listing_Price    12000000
Name: 3, dtype: object
```

10. Data Visualization Using Matplotlib

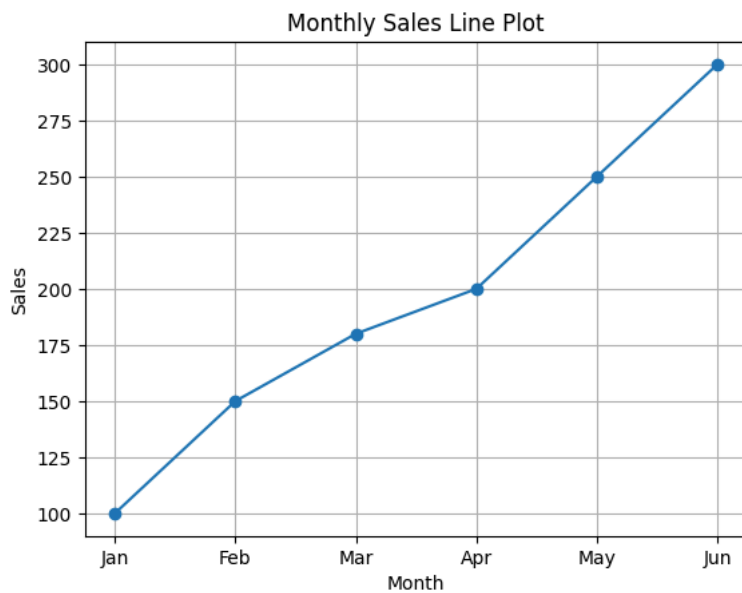
```
import matplotlib.pyplot as plt
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [12000, 18000, 15000, 22000, 25000, 27000]
plt.figure(figsize=(6,4))
plt.plot(months, sales, marker='o')
plt.title("Monthly Sales - Line Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.grid(True)
plt.show()
plt.figure(figsize=(6,4))
plt.bar(months, sales)
plt.title("Monthly Sales - Bar Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```



11. Sales Prediction Visualization Using Matplotlib

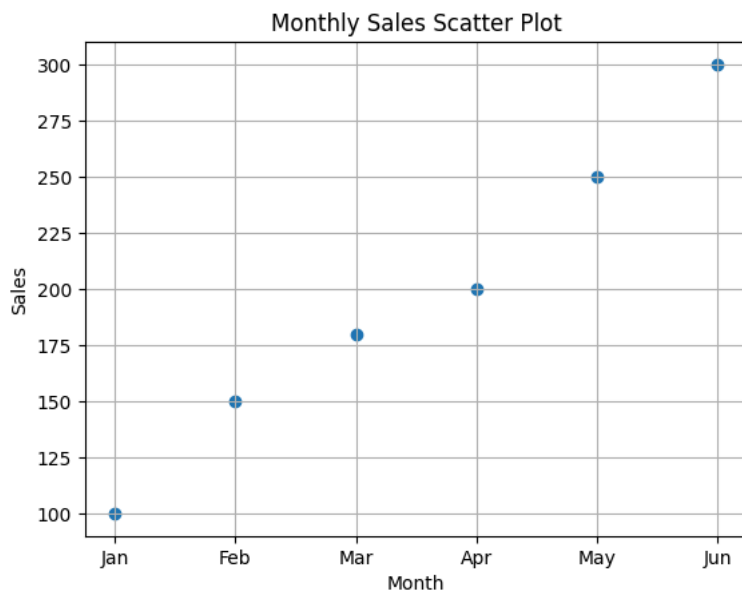
1.Line Plot

```
import matplotlib.pyplot as plt
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [100, 150, 180, 200, 250, 300]
plt.plot(months, sales, marker='o')
plt.title("Monthly Sales Line Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid(True)
plt.show()
```



2.Scatter Plot

```
import matplotlib.pyplot as plt
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [100, 150, 180, 200, 250, 300]
plt.scatter(months, sales)
plt.title("Monthly Sales Scatter Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid(True)
plt.show()
```



3. Bar Plot

```
import matplotlib.pyplot as plt
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [100, 150, 180, 200, 250, 300]
plt.bar(months, sales)
plt.title("Monthly Sales Bar Plot")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

